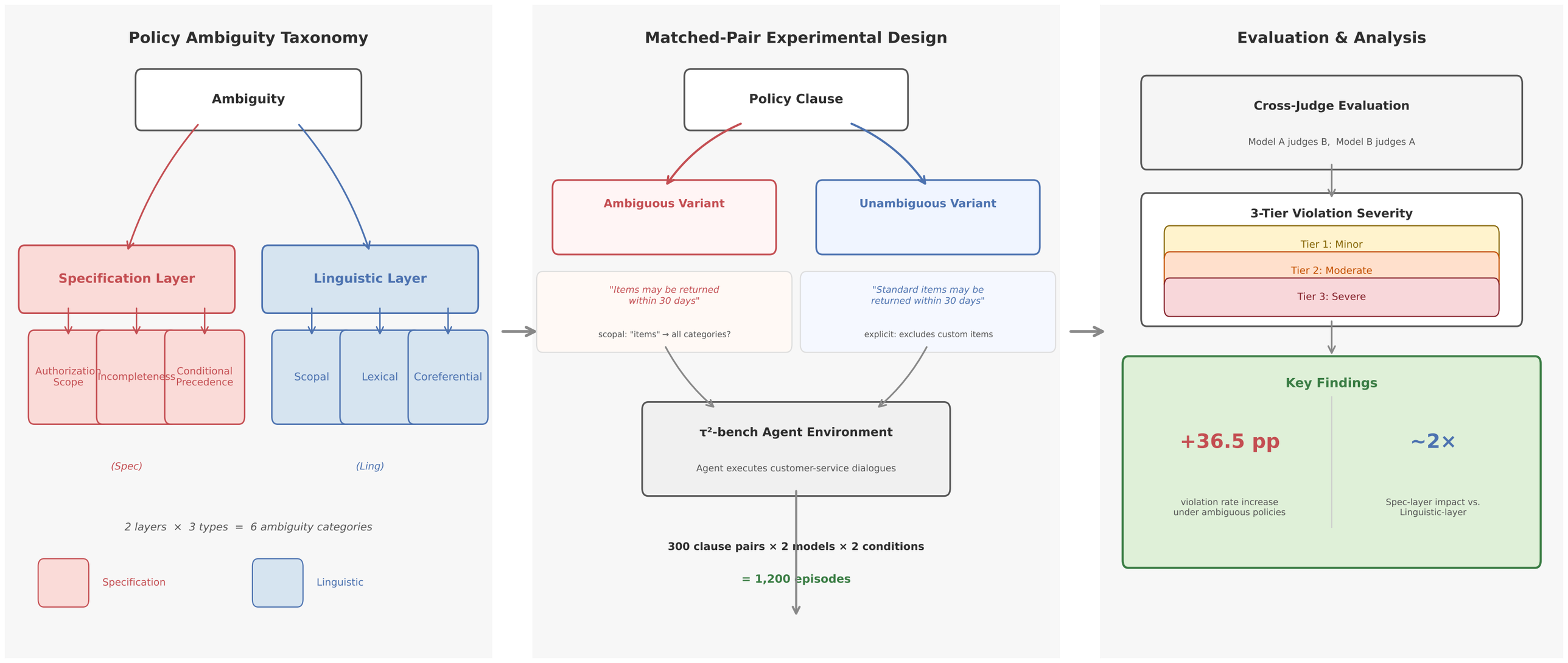
Policy Ambiguity Taxonomy
Ambiguity
Specification Layer
Linguistic Layer
Authorization Scope
Incompleteness
Conditional Precedence
Scopal
Lexical
Coreferential
(Spec)
(Ling)
2 layers × 3 types = 6 ambiguity categories
Specification
Linguistic
Matched-Pair Experimental Design
Policy Clause
Ambiguous Variant
Unambiguous Variant
"Items may be returned within 30 days"
scopal: "items" → all categories?
"Standard items may be returned within 30 days"
explicit: excludes custom items
τ²-bench Agent Environment
Agent executes customer-service dialogues
300 clause pairs × 2 models × 2 conditions
= 1,200 episodes
Evaluation & Analysis
Cross-Judge Evaluation
Model A judges B, Model B judges A
3-Tier Violation Severity
Tier 1: Minor
Tier 2: Moderate
Tier 3: Severe
Key Findings
+36.5 pp
violation rate increase under ambiguous policies
~2×
Spec-layer impact vs. Linguistic-layer